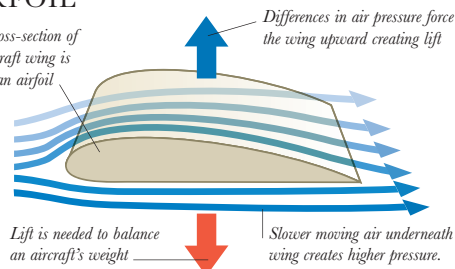


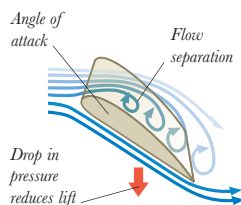
AIRFOIL

The cross-section of an aircraft wing is called an airfoil.



LIFT

As the curved wing moves through the air, the air passing over the wing moves faster than the air passing beneath. Fast-moving air has a lower pressure, so slower, high-pressure air beneath the wing forces it upward.



STALLING

Stalling (when an aircraft drops suddenly) occurs when the critical angle of attack between the wing's centerline and the airflow direction is exceeded. This causes a breakdown in airflow, leading to a drop in pressure and reducing lift.

Elevator

A control surface on the tailplane's trailing edge, which governs pitch.

Elevon

Wing control surface – mostly found on delta-wing aircraft – that combines the functions of elevator and aileron.

Envelope

The part of an airship that contains the lifting gas.

Fairing

A covering used to streamline any projections in the aircraft structure.

Fin

The vertical section of the tail or wingtip, used to increase the stability of the aircraft. Also known as a vertical stabilizer.

Flight deck

The compartment housing the crew and flight controls in a large aircraft or spacecraft.

Flight simulator

A training device that creates a simulated flying environment and experience within a virtual cockpit.

Fly-by-wire

An electronic flight control system used instead of mechanical controls.

Flying boat

An aircraft with a watertight fuselage, resembling the shape of a boat's hull, allowing it to take off and land in water.

Flying wing

An all-wing aircraft without the

customary fuselage, and a tail designed to be aerodynamically efficient.

Fuselage

The main body of an aircraft, to which the wings and tail are attached.

G-force

The force experienced by the crew of a craft undergoing rapid acceleration or deceleration. One "g" is equal to the force normally exerted by Earth's gravity.

Glider

A light aircraft without an engine.

Gondola

The passenger compartment that hangs from a balloon or airship.

Hangar

An enclosed structure with a large floor area used to house aircraft.

Hang glider

A fabric sailing wing attached to a light, triangular frame, beneath which the pilot hangs horizontally.

Head-up display (HUD)

A unit that projects information, such as combat status and aircraft performance data, onto a transparent screen in the pilot's line of sight, lessening the need to look down into the cockpit.

Helicopter

See HELICOPTER panel, p. 431.

Horsepower

A unit of power devised by the 19th-century British engineer James Watt, which describes the

energy required to raise 550lb (250kg) one foot (30cm) in one second.

HOTAS

HOTAS (Hands On Throttle and Stick) describes a system whereby the switches are positioned on the throttle and the control stick so that the pilot does not have to remove his or her hands from them.

Hypersonic

A flight speed equal to or greater than Mach 5 (i.e. five times the speed of sound).

Hypoxia

An oxygen deficiency in the body tissues, which can be brought on by flying at high altitudes (where there is less oxygen) in an unpressurized environment.

INS

INS (Inertial Navigation System) is a cockpit device that provides positional and navigational information without the need for data from external references.

Interrupter gear

A device that allowed a fixed machine gun to fire through a propeller, by halting the gun's fire whenever a blade was in line with the barrel of the gun.

Jet-stream

A high-altitude, high-speed wind flow stream.

J-STARS

Joint-STARS (Surveillance Target Attack Radar System) is a battle-management surveillance system carried in a modified Boeing airliner (E-8), which detects, tracks, and targets enemy activity on the ground, relaying the information to US Air Force and US Army command posts.

Landing gear

see Undercarriage.

Leading edge

The front, rounded edge of an aerofoil such as a wing or rotor.

Lift

The force exerted on a moving aerofoil that causes a wing to rise. See also AIRFOIL panel, above.

Longeron

Any main load-bearing fuselage structural member with a fore and aft axis.

Look-down Shoot-down

The capability of an aircraft flying at high altitude to use advanced radar systems to detect a fast, low-flying enemy aircraft against a jumble of radar reflections from the ground.

Mach number

The ratio of an aircraft's speed to

the speed of sound. Mach 2, for example, is twice the speed of sound.

MFD (Multifunction Display)

A cockpit instrument able to display two or more types of information, such as navigational and fuel consumption data.

Monoplane

Airplane with one set of wings. See also ANATOMY OF AN AIRPLANE panel, left.

Monocoque

A structure, such as a fuselage, that has no internal bracing and derives most of its strength from its skin.

NACA

NACA (National Advisory Committee on Aeronautics) was founded during WWI and renamed NASA in 1958.

NACA Cowling

A ring-shaped cowling of a certain design, used to cover radial engines and reduce drag.

Nacelle

A streamlined, protective enclosure housing a part of the aircraft likely to induce drag, such as engines, crew, or armament.

NASA

Founded in 1958, NASA (National Aeronautics and Space Administration) is the US government agency that conducts all nonmilitary spaceflight and aeronautical research.

Ornithopter

An aircraft with oscillating wings.

Parachute

A fabric canopy that allows a person or package to descend safely to earth.

Payload

The load carried by an aircraft, including passengers and cargo, from which revenue is obtained.

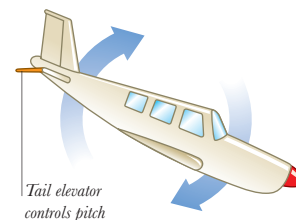
Pilot

The person with primary responsibility for flying an aircraft.

Piston engine

See PISTON ENGINES panel, p. 430.

BASIC CONTROLS

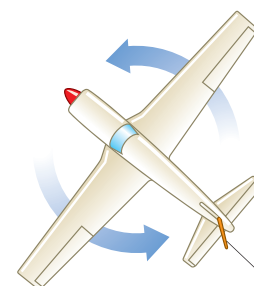
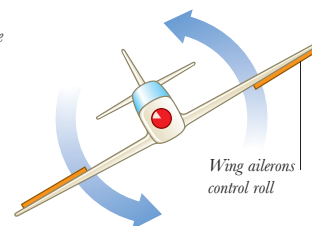


PITCH

To pitch (climb or dive), the pilot pushes or pulls the control column, raising or lowering the elevator flaps on the tailplane. This causes movement along the aircraft's lateral axis.

ROLL

To roll, the pilot moves the control column to the left or right, which raises the ailerons on one wing and lowers them on the other. This causes movement along the aircraft's longitudinal axis.



YAW

To yaw left or right, the pilot's feet swivel the rudder bar, turning the upright rudder on the airplane's fin. This causes movement along the aircraft's normal axis.